

The cognitive turn in computational modeling for experimental pragmatics

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Abstract

concrete.

1 Experimental pragmatics and computational modeling

The linguistic subfield of pragmatics studies how language is used for efficient communication and how communicated meaning is modulated, if not created, in a particular context by systematic, context-general patterns of use. Its origins lie in the Philosophy of Language, in particular the analytical tradition of philosophy, in work on topics like *speech-acts* (Austin, 1962; Searle, 1969), *conversational implicature* (Grice, 1975), or *presupposition* (Stalnaker, 1973; Karttunen, 1973). Much early work on these topics remained invested in a logical-formal approach (Gazdar, 1979; Kadmon, 2001), but the early 2000's brought a deep and empowering incision: a widely adopted and ultimately very successful *experimental turn* (Noveck and Sperber, 2004; Noveck, 2018), in parallel to similar empirical re-orientations in other hitherto very analytically minded subfields of linguistics like formal syntax (Sprouse, 2007, 2023) and formal semantics (Meibauer and Steinbach, 2011; Cummins and Katsos, 2019).

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- modeling in general:
 - all models are wrong; some are useful
 - even if all models are wrong, we should prefer models that are less wrong in the sense of Isaak
 - so how our (less) wrong models useful?
 - while for data: garbage in — garbage out
 - for models: more assumptions in — more conclusions out
 - e.g., causal modeling
 - reasonable assumptions in; interesting conclusions out
 - all models contain some assumptions about the **data-generating process**

- more assumptions about the nature of that process → more information gained from the data
- two specific contributions of explicit (computational / formal) modeling in XPrag:
 1. **bridge to linguistic theory:** (a successful past)
 - give linguistic theory (e.g., latent concepts) an explicit role in our picture of the data-generating process via theory-driven bespoke modeling
 - experimental turn: experiments based on theory
 - bring in the theory as a first-order citizen → theory-driven statistical modeling
 2. **bridge to general cognition:** (a possible (bright) future)
 - frame pragmatic reasoning in the broader picture of human cognition
 - stuff for the future: proclaiming the *cognitive turn* in experimental pragmatics

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